Here we collect some basic facts relating independence of random variables with their characteristic functions. The main result is the following, whose proof requires two more theorems proved below.

Teorema 1. *Let* $X = (X_1, .., X_n)$ *and suppose that* $\forall \xi \in \mathbb{R}^n$ $\phi_X(\xi) = \prod_{j=1,..,n} \phi_{X_j}(\xi_j)$. *Then* $X_1, .., X_n$ *are independent.*

Proof. Let $Y = (Y_1, .., Y_n)$ vector of independent random variables, with $Y_j \sim X_j$ for $j = 1, .., n$. Now we prove $\phi_X = \phi_Y$, which will give us the thesis applying the unicity theorem.

$$\begin{aligned}\phi_Y(\xi) =&\mathbb{E}[exp(i \cdot \sum_{j=1,..,n} \xi_j Y_j)] = \mathbb{E}[\prod_{j=1,..,n} exp(i\xi_j Y_j)] \\ &= \prod_{j=1,..,n} \mathbb{E}[exp(i\xi_j Y_j)] = \prod_{j=1,..,n} \phi_{Y_j}(\xi_j) = \prod_{j=1,..,n} \phi_{X_j}(\xi_j)\end{aligned}$$

By the unicity theorem we get [1]. $F_X = F_Y$, so in particular X_j's must be independent since the Y_j's are.

□

Teorema 2. *Let* $X, Y : (\Omega, \mathcal{A}) \to (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$. *Suppose* $\phi_X = \phi_Y$. *Then* $F_X = F_Y$.

[1] for any d-dimensional random varibale X, define $F_X : \mathbb{R}^d \to [0, 1], (x_1, .., x_d) \mapsto \mathbb{P}(X_1 \leq x_1, .., X_d \leq x_d)$